

```
In [1]: print(40 + 45.4)
```

85.4

```
In [2]: print(int(40 + 45.5))
```

85

```
In [3]: print(50 - 23)
print(23 - 42)
```

27

-19

```
In [4]: print(-(23-34))
```

11

```
In [5]: print(23-45.7)
```

-22.700000000000003

```
In [6]: print(int(23-45.7))
```

-22

```
In [7]: print(23*34.45)
```

792.35

```
In [12]: print(int(23*34.45))
print(23**34.45)
```

792

8.156877700688407e+46

```
In [9]: print(float(45*56))
```

2520.0

```
In [10]: print(34/31)
```

1.096774193548387

```
In [11]: print(34//31)
```

1

```
In [13]: print(34%31)
```

3

```
In [14]: print(float(34%31))
```

3.0

```
In [1]: print(type(12))
print(type(123.34))
print(type(23+3j))
print(type("yokoso..."))
print(type([2,3,4,5,6,7,45,33,23,123]))
print(type({'name': 'bharadwaj'}))
print(type({34,3}))
```

```
<class 'int'>
<class 'float'>
<class 'complex'>
<class 'str'>
<class 'list'>
<class 'dict'>
<class 'set'>
```

```
In [2]: print('yokoso, watashi no soul society ye!')
```

```
yokoso, watashi no soul society ye!
```

```
In [3]: 'yokoso, watashi no soul society ye!'
```

```
Out[3]: 'yokoso, watashi no soul society ye!'
```

```
In [4]: print('welcome to aizen's soul society')
```

```
Cell In[4], line 1
    print('welcome to aizen's soul society')
                                ^
SyntaxError: unterminated string literal (detected at line 1)
```

```
In [5]: print('welcome to aizen\'s soul society')
```

```
welcome to aizen's soul society
```

```
In [6]: print('watashi'+ 'no soul society')
```

```
watashino soul society
```

```
In [7]: print('watashi' + ' no soul society')
```

```
watashi no soul society
```

```
In [8]: 'yokoso ' * 5
```

```
Out[8]: 'yokoso yokoso yokoso yokoso yokoso '
```

```
In [11]: print('watashi no\nsoul society')
```

```
watashi no
soul society
```

```
In [13]: print('yokoso,\nwatashi no\nsoul society')
```

```
yokoso,
watashi no
soul society
```

```
In [14]: x=10
          x
```

```
Out[14]: 10
```

```
In [16]: x+12
```

```
Out[16]: 22
```

```
In [17]: y=13
          y
```

Out[17]: 13

In [18]: `x+y`

Out[18]: 23

In [19]: `x`

Out[19]: 10

In [20]: `x+12`

Out[20]: 22

In [21]: `x`

Out[21]: 10

In [22]: `_+x`

Out[22]: 20

In [23]: `_+34`

Out[23]: 54

In [25]: `name= 'bharadwaj'`
`name+ ' has bankai'`

Out[25]: 'bharadwaj has bankai'

In [26]: `len(name)`

Out[26]: 9

In [27]: `name[0]`

Out[27]: 'b'

In [28]: `name[5]`

Out[28]: 'd'

In [29]: `name[7]`

Out[29]: 'a'

Operators

arithmetic operators

```
In [30]: x1,y1=3,9  
print(x1+y1)
```

12

```
In [31]: x1-y1
```

Out[31]: -6

```
In [32]: x1*y1
```

Out[32]: 27

```
In [33]: y1/x1
```

Out[33]: 3.0

```
In [34]: y1//x1
```

Out[34]: 3

```
In [35]: y1**x1
```

Out[35]: 729

```
In [36]: y1%x1
```

Out[36]: 0

assignment operators

```
In [37]: x =2  
x=x+2
```

```
In [38]: x
```

Out[38]: 4

```
In [39]: x+=2
```

```
In [40]: x
```

Out[40]: 6

```
In [41]: x+=2  
x
```

Out[41]: 8

```
In [42]: x+=2  
x
```

Out[42]: 10

```
In [43]: x+=2  
x
```

Out[43]: 12

```
In [44]: x-=2  
x
```

Out[44]: 10

```
In [45]: x-=2  
x
```

Out[45]: 8

```
In [46]: x-=2  
x
```

Out[46]: 6

```
In [47]: x*=2  
x
```

Out[47]: 12

```
In [49]: x*=2  
x
```

Out[49]: 20

```
In [50]: x*=2  
x
```

Out[50]: 40

```
In [51]: x*=2  
x
```

Out[51]: 80

```
In [52]: x/=2  
x
```

Out[52]: 40.0

```
In [53]: x/=2  
x
```

Out[53]: 20.0

```
In [54]: x/=2  
x
```

Out[54]: 10.0

```
In [55]: x//=2  
x
```

```
Out[55]: 5.0
```

```
In [56]: x//=2  
x
```

```
Out[56]: 2.0
```

unary operator

```
In [57]: n=19  
n
```

```
Out[57]: 19
```

```
In [58]: m=-(n)  
m
```

```
Out[58]: -19
```

relational operators

```
In [59]: a=12  
b=67  
a==b
```

```
Out[59]: False
```

```
In [60]: a>b
```

```
Out[60]: False
```

```
In [61]: a<b
```

```
Out[61]: True
```

```
In [62]: a!=b
```

```
Out[62]: True
```

logical operators

```
In [64]: a>2 and b>100
```

```
Out[64]: False
```

```
In [65]: a>2 or b>100
```

Out[65]: True

number system conversion

In [67]: `bin(12)`

Out[67]: '0b1100'

In [68]: `int('0b1100')`

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[68], line 1  
----> 1 int( )  
  
ValueError: invalid literal for int() with base 10: '0b1100'
```

In [69]: `int('0b1101',2)`

Out[69]: 13

In [70]: `int(1100)`

Out[70]: 1100

In [71]: `int(ob1100)`

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[71], line 1  
----> 1 int(ob1100)  
  
NameError: name 'ob1100' is not defined
```

In [72]: `int(0b1100, 2)`

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[72], line 1  
----> 1 int(0b1100, 2)  
  
TypeError: int() can't convert non-string with explicit base
```

In [73]: `0b1100`

Out[73]: 12

In [74]: `int(0b1101)`

Out[74]: 13

In [75]: `oct(65)`

Out[75]: '0o101'

In [76]: `int(0o101)`

Out[76]: 65

In [77]: `hex(126)`

Out[77]: '0x7e'

In [78]: `0zx7e`

Cell In[78], line 1
`0zx7e`
`^`
SyntaxError: invalid decimal literal

In [81]: `int(0x7e)`

Out[81]: 126

swapping variables

In [83]: `a=4`
`b=56`

`a=b`
`b=a`
`a,b`

Out[83]: (56, 56)

In [84]: `a=4`
`b=56`
`a,b = b,a`
`print(a,b)`

56 4

In [85]: `a1=56`
`b1=86`
`temp = a1`
`a1=b1`
`b1=temp`
`print(a1,b1)`

86 56

In [86]: `a1 = a1 + b1`
`b1 = a1 - b1`
`a1 = a1 - b1`
`print(a1,b1)`

56 86

In [87]: `a1, b1 = b1, a1`
`print(a1, b1)`

86 56

Bitwise Operators

compliment opearator

```
In [90]: ~19
```

```
Out[90]: -20
```

```
In [91]: ~192
```

```
Out[91]: -193
```

```
In [92]: ~~193
```

```
Out[92]: 193
```

```
In [93]: ~-193
```

```
Out[93]: 192
```

bitwise '&', '|' and '^'

```
In [97]: 12 & 15
```

```
Out[97]: 12
```

```
In [98]: 12 & 25
```

```
Out[98]: 8
```

```
In [99]: 12 | 15
```

```
Out[99]: 15
```

```
In [100... 12 | 25
```

```
Out[100... 29
```

```
In [101... 12 ^ 15
```

```
Out[101... 3
```

```
In [102... 12 ^ 25
```

```
Out[102... 21
```

bitwise right shift '>>' and left shift '<<'

In [104... `20<<4`

Out[104... 320

In [105... `12<<4`

Out[105... 192

In [106... `10<<2`

Out[106... 40

In [107... `10>>2`

Out[107... 2

In [108... `20>>4`

Out[108... 1

In [109... `20>>2`

Out[109... 5

import math module

In [111... `x = sqrt(4)`

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[111], line 1  
----> 1 x = sqrt(4)  
  
NameError: name 'sqrt' is not defined
```

In [113... `import math`
`x = math.sqrt(4)`
`x`

Out[113... 2.0

In [114... `x= math.cbrt(8)`

In [115... `x`

Out[115... 2.0

In [117... `x= math.pow(2, 3)`
`x`

Out[117... 8.0

In [118... `x = math.factorial(12)`
`x`

Out[118... 479001600

```
In [120... print(math.floor(12.34))
print(math.ceil(12.34))
print(math.trunc(12.34))
print(math.fabs(-12.34))
```

12
13
12
12.34

```
In [121... print(math.pi)
```

3.141592653589793

```
In [122... print(int(math.pi))
```

3

```
In [124... from math import sqrt, pow, factorial, floor, ceil, trunc, fabs, pi
pow(2, 3)
```

Out[124... 8.0

```
In [125... round(pow(2, 3))
```

Out[125... 8

user input function

```
In [127... a = input()
b = input()
c = a + b
print(c)
```

245

```
In [132... a = input('Enter first number: ')
a1= int(a)
b = input('Enter second number: ')
b1 = int(b)
c=a1+b1
c
```

Out[132... 7

```
In [133... ch = input('Enter a character: ')
print(ch)
```

welcome to my world

```
In [134... print(ch[3])
```

c

```
In [135... ch = input('Enter a character: ')[1:5]
print(ch)
```

elco

In []: